LLM-autocorprophagy

Predictions, retrodictions and prognoses of LLMs' impact on written language.

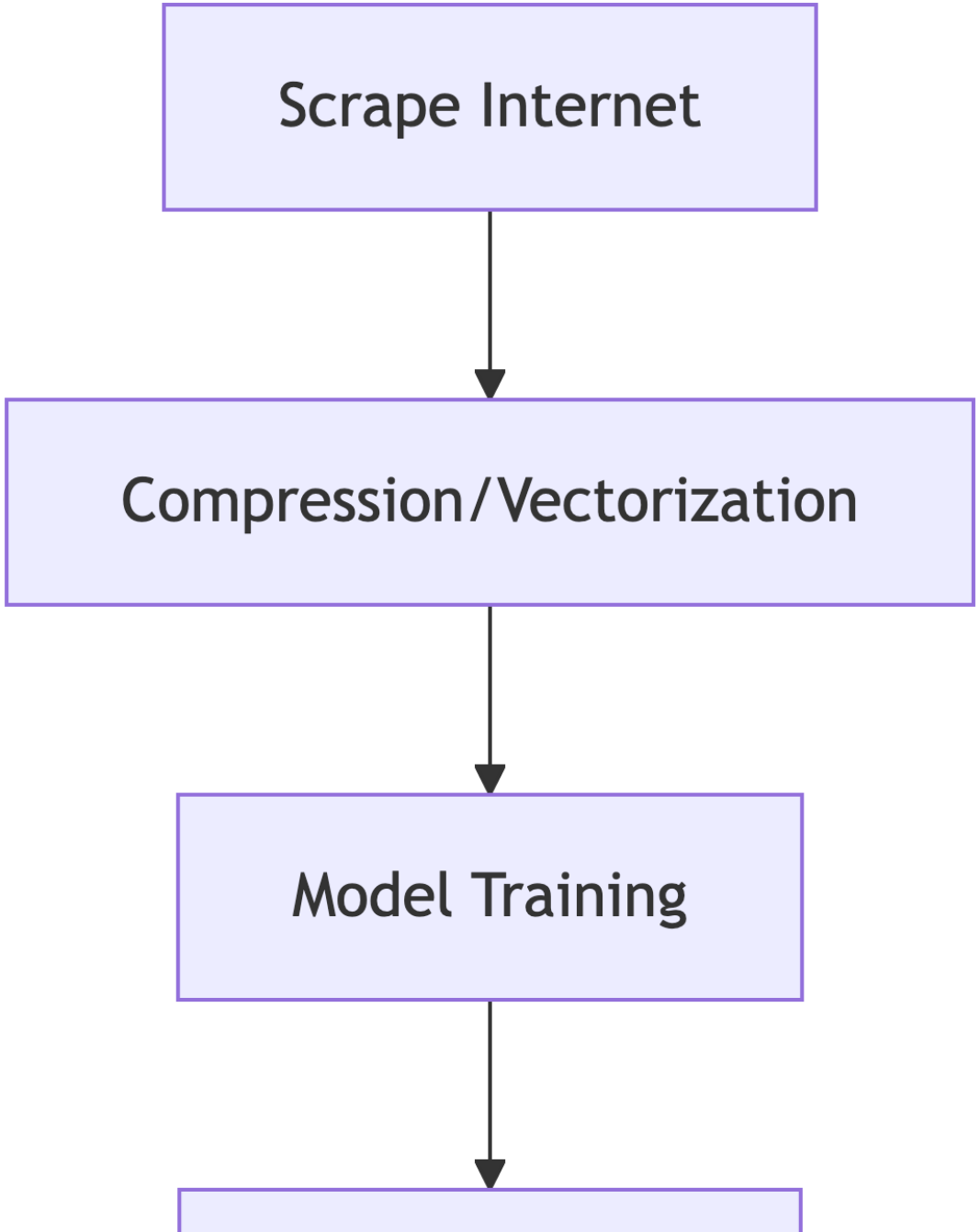
Aleksander Dietrichson, PhD

Disclaimer & Reverse Disclaimer

- *Claude code* was used to generate the source code for data collection and for running the experiment as reported.
- No AI tools were used in the creation of this presentation. R and quarto code for data analysis written exclusively written by the author[1].
- Source-code, AI prompts and data available at: https://github.com/dietrichson/ILA-2026

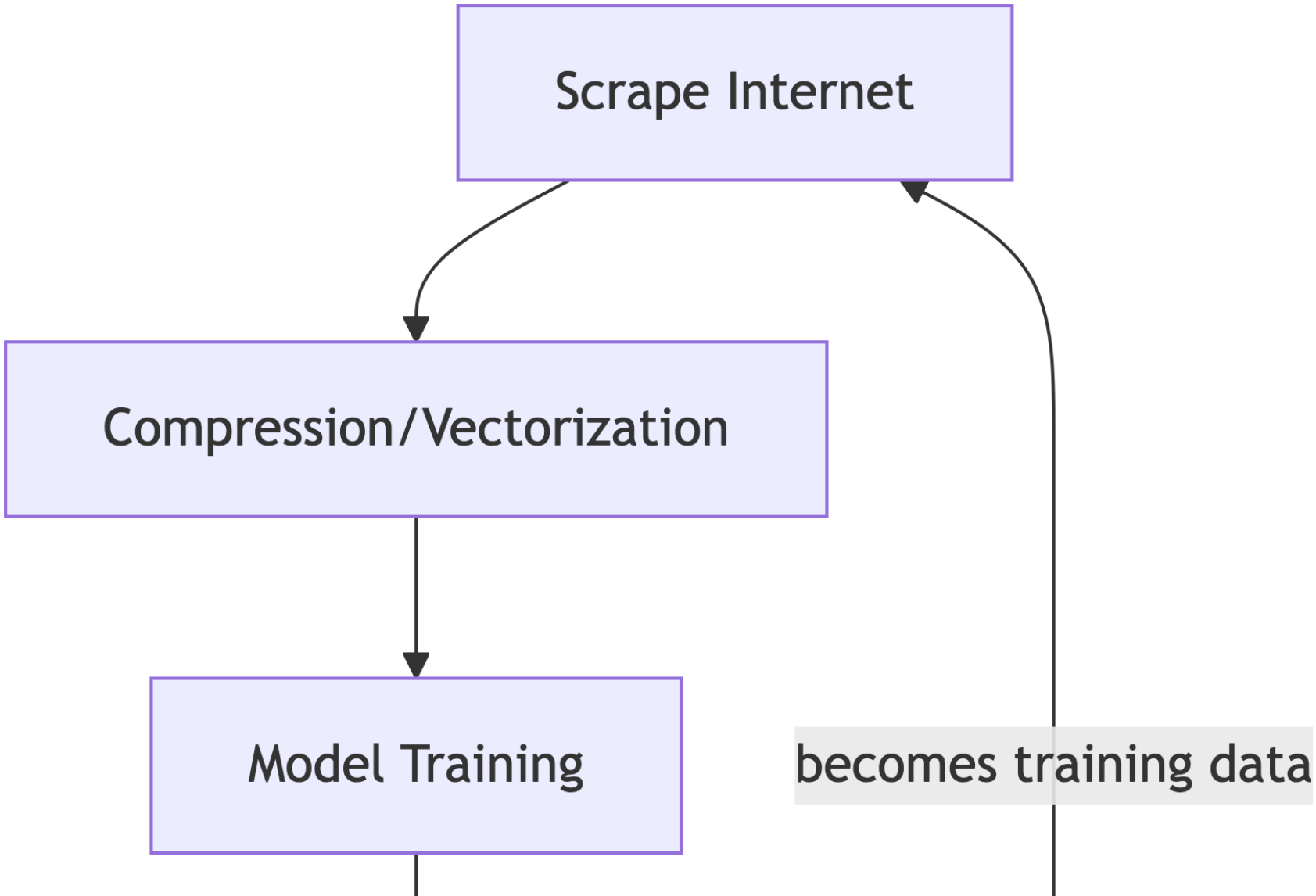
[1] With the exception of spell-checker.

How LLMs Work



The Problem

When most training data for LLMs is the output of other LLMs a loop is created



Experiment: Data

- Five articles, 1600-2200 words each.
- One from the *New Yorker* four from the *New York Times*.
- Chosen for length, from the latest edition of each publication (last week of april 2026)

Experiment: Method

Step 1: Summarization task

This is the text of an published article. Your task is to summarize this. Aim for 500 words. The output filename should have the same name as the input-file with "_summary" interjected.

Step 2: Expansion task

This is a short article, we would like to make it longer. Please add verbiage so that the article is around 1600 words long. The output filename should be the same as the input-file with "_expanded" instead of "_summary"

Experiment: Aparatus

- Claude code was used, with Opus 4.7

Results

Table 1: Unique words (types) per condition

Publication	Original	Processed	Loss
New Yorker	792	753	4.9%
New York Times 1	818	690	15.6%
New York Times 2	992	741	25.3%
New York Times 3	820	640	22.0%
New York Times 4	921	677	26.5%

Average loss is 19%

Results

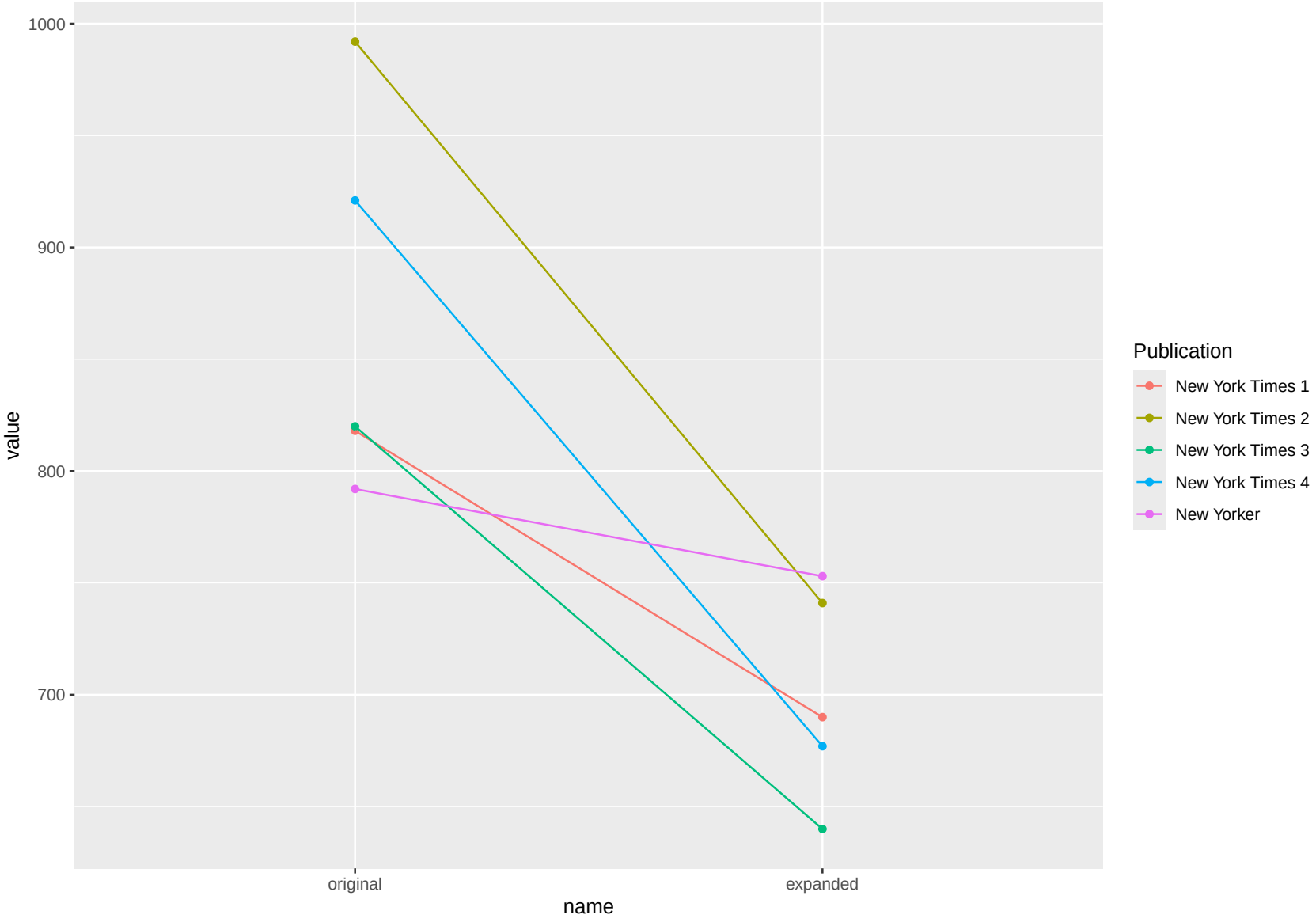


Figure 1: Unique words (types) per condition

Empirical Research

- Random sample of 15 thousand webpages
- Retrieved from https://commoncrawl.org/
- English language
- 1000 from each of the years 2012 through 2026
- Calculated Type/Token ratio for each document

Type Token Ratio

$$TTR = \frac{\text{number of unique words}}{\text{total number of words}}$$

A measure of *Vocabulary diversity* in text

Results

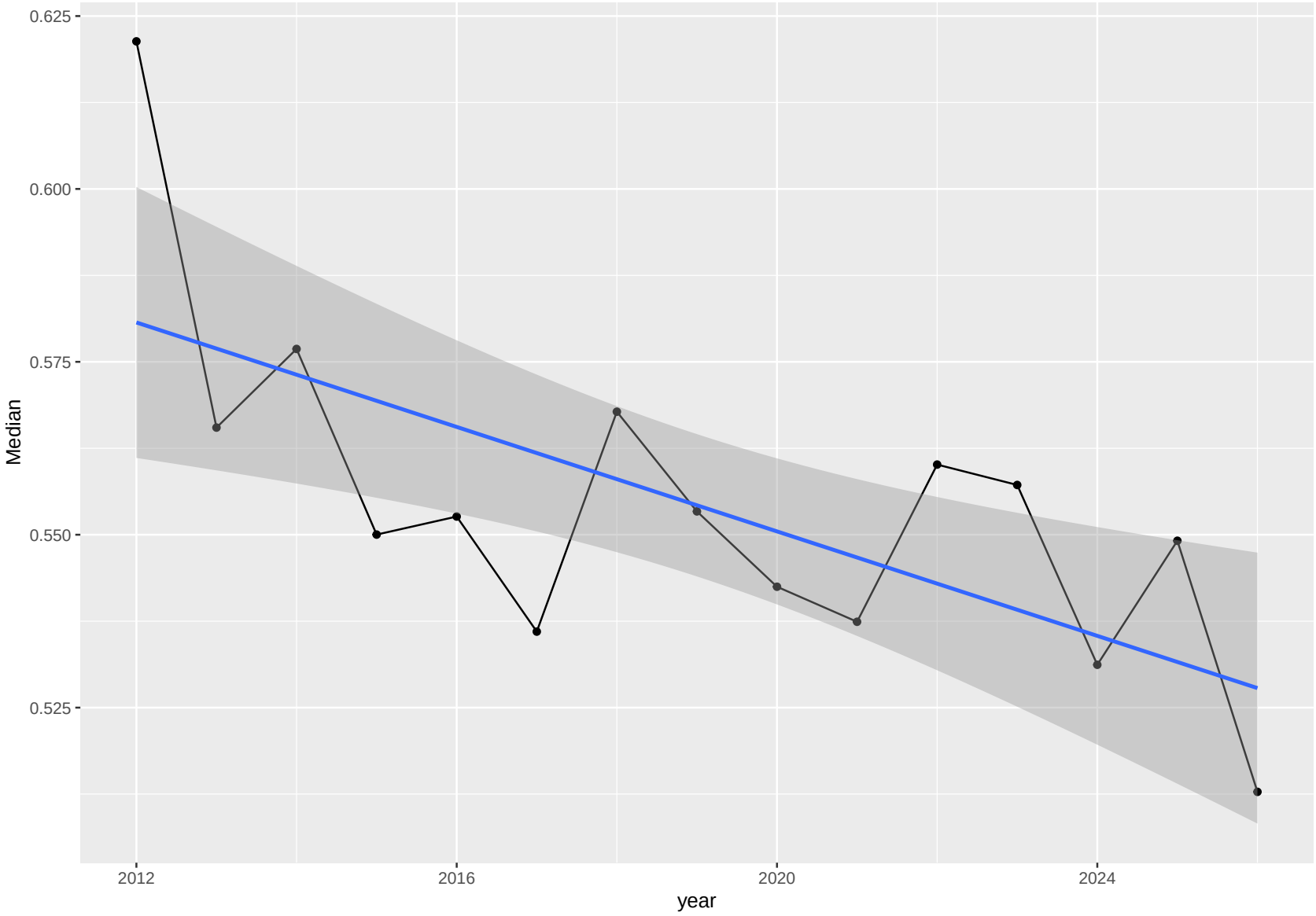


Figure 2: Median TRR over Year

Results

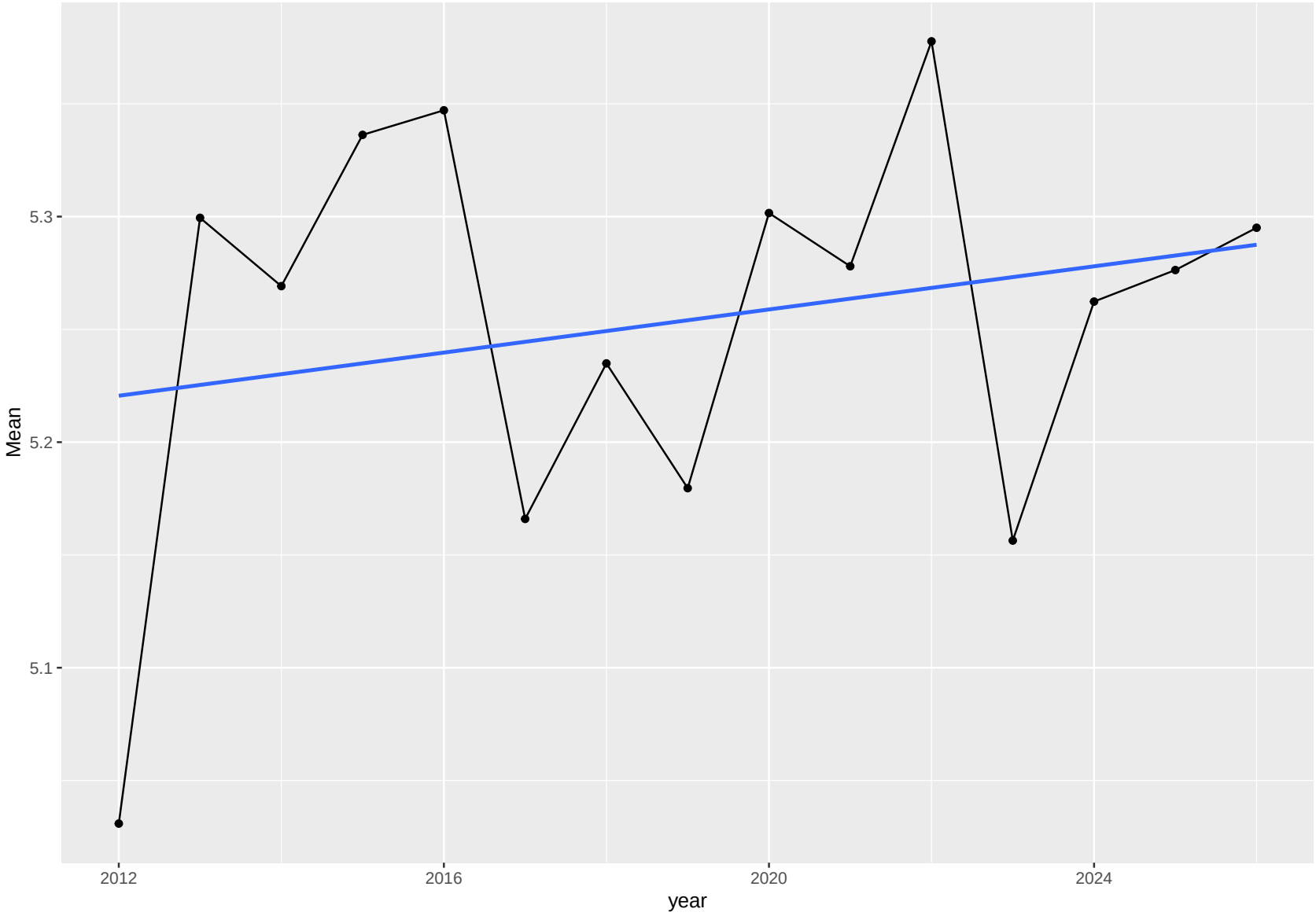


Figure 3: Mean Length of Words Over Year

Results

- Type/Token ratio drops by 0.0077516 ($\sim$ .8%) anually.
- Word-length increases by 0.0188675 character per word anually.

Projections

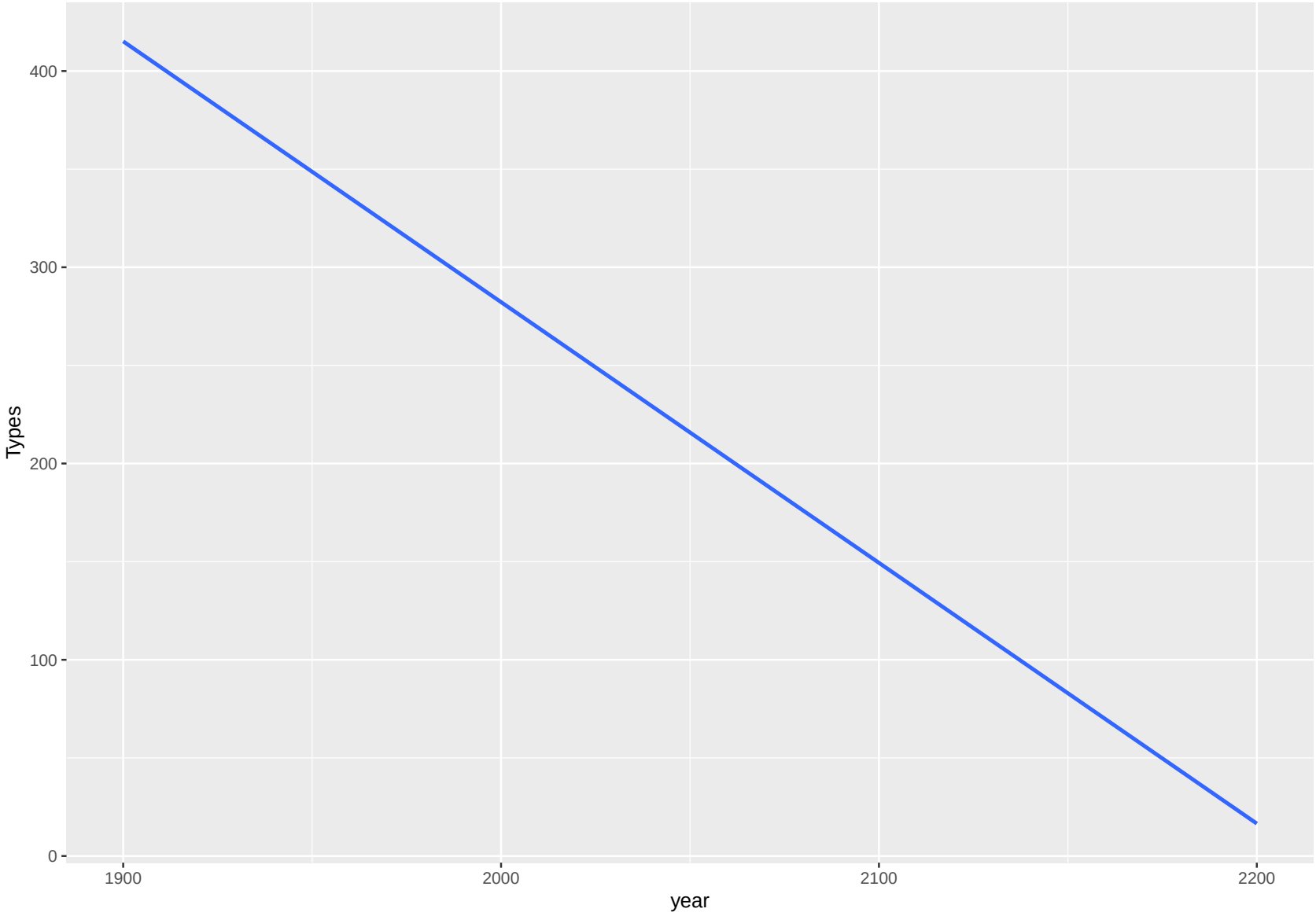


Figure 4: Mean Types (Raw Scores)

Conclusions

- Linguistic diversity is under preassure.
- "AI slop" seems omnipresent and still feature recognizable characteristics